

# Meta-modeling Business Architecture

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## Summary

This whitepaper sets forth the case for creating and standardizing a meta-model for Business Architecture. It defines a set of basic principles to guide the creation of a Business Architecture meta-model.

## On the need for a Business Architecture Meta-model

### What a meta-model does

A model is a network of data that represents a reality. The information a bank keeps about accounts is a model. The reality is that the bank has a pile of cash stored in a vault, but the cash is not bound up in packets that are labeled with account numbers. Instead, the bank keeps track in a model of how much of the cash belongs to each account. Models are easier to manipulate than is the corresponding reality. In the case of the bank, manipulating the model in an IT system is much quicker and less expensive than hiring people to go into the vault and move money from packet to packet.

To be useful, a model must represent some consensus view of reality. It would not work if the bank had multiple accounting models, each representing someone's view of how the pile of cash should be allocated across the accounts. One of the most challenging aspects of model building is determining this consensus view of the reality.

Business architecture as a discipline emerged as a synthesis of a number of independent ideas about modeling businesses. One of these ideas is that businesses of a certain type (e.g. retail banks) have a common generic functional structure; in business architecture, this structure is represented in a Capability Map. Another idea is a simplification of the processes by which a business creates value for its clients. A third idea is the linkage of capabilities and value stages. Models created from these ideas have had idiosyncratic variations, depending on the knowledge and experience of the business architect creating them. However, the community of business architects is now several thousand in number and inconsistency in modeling practices inhibits the sharing of knowledge and the development of information tools to support the practice and make it more efficient.

A meta-model is a recipe for creating models. The meta-model specifies the elements that can be used when creating models and it specifies the relationships that can be created between those elements. For example, a meta-model for business architecture will specify model elements that are capabilities, value stages and value streams. The meta-model will also specify that a value stream consists of a sequence of value stages and that a value stage is related to a number of capabilities that are needed to create the value produced in the value stage. Having a meta-model insures that models are created in a consistent way.

In addition, a meta-model is a template for practitioners and architecture tools vendors who wish to build tools to create models and communicate them to decision makers and others. Because a model is a tangible thing, it can be shared through technology. This sharing leads to opportunistic learning and it permits feedback that results in improving both the models created from the meta-model and improving the meta-model itself.

## Benefits and opportunities for meta-model standardization

Standardization is a process whose end result is a consensus definition of a product or service. A good standardization process will include the interests of the primary stakeholders and providers, and not favor any particular provider over others.

Standardization is possible to the extent that a consensus can be found. This generally rules out standardizing experimental products and services. It also rules out products and services in environments that are highly competitive and where product or service differentiation is a primary component of delivered value.

Standardization can occur as a result of a formal process, carried out by a recognized standardization organization. It can also occur as the result of a group of market participants acting together to capture the largest share of the market with a common product or service.

The OMG is the preeminent standardization organization for design and architecture meta-models. It has standardized meta-models for UML and BPMN and has a number of business-related meta-model standardization processes underway. The Business Architecture Guild has worked cooperatively with the OMG since its beginning and has recently entered into collaborative agreements with the OMG. Consequently, the OMG would be the first choice of the Guild to standardize a Business Architecture meta-model

## Business Architecture meta-model principles

### Meta-model scope and model integration

Business Architecture co-exists with many other business-oriented methods. Part 2 of the **Guide to the Business Architecture Body of Knowledge** defines the core components of the Business Architecture ecosystem and explains how these components relate to other practices such as strategy, organization structure, initiatives and business models. In addition, the Business Architecture Guild has recently published a whitepaper aligning Business Architecture with business process management ([Business Architecture and BPM – Differentiation and Reconciliation](#)).

Business Architecture captures a point of view on an enterprise that emphasizes value, functional structure and transparency. Other methods capture other points of view. Strategy, for example, focuses on the enterprise's relationship to its competitors, customer and suppliers. BPM focuses on the enterprise's operational behavior and efficiency.

The core blueprints of Business Architecture, Capability Mapping, Value Mapping and Information Mapping are stable models that apply without change to all businesses of a particular type, regardless of the business model, strategy, organization structure, operational behavior model or management model used by each particular business. This characteristic of Business Architecture makes it possible to relate specific business models, strategies, organization structures, operational behavior models and management models to the Business Architecture and also to each other. The Business Architecture point of view permits the integration of other points of view and consequently facilitates analyses that correlate multiple points of view.

As the primary purpose of a meta-model is to define the types of models that can be created, the principles defined in the preceding paragraphs have two consequences for the definition of the Business Architecture meta-model:

1. The Business Architecture meta-model must be capable of relating its elements to the meta-model elements used to capture other points of view on businesses;
2. The Business Architecture meta-model, when related to another meta-model (e.g. for strategy), must not conflict with the related meta-model. In practice, this means that where the Business Architecture meta-model contains elements specifically intended to be used in relations to other meta-models, these Business Architecture meta-model elements must be defined in the most generic way. In other words, Business Architecture does not introduce a new way to express and analyze strategy (for example); instead, it provides a few meta-model elements that can be used to relate a strategy method (e.g. Strategy Maps, Five Forces) to Capability Maps, Value Maps and Information Maps.

The scope of the Business Architecture meta-model is thus defined by the other points of view that are desirable to integrate with Business Architecture. The meta-model itself must be architected to allow integration of Business Architecture models with other models.

### Extensibility of the meta-model

Business Architecture is a new method and point of view, whose core components are stable and suited for standardization. However, other components are still maturing and new components may be introduced as the Business Architecture method and other points of view evolve. For this reason, the Business Architecture meta-model must allow for future extension and it must not overly constrain the kinds of models that can be created from it.

Consequently, the meta-model may allow models to be created that are outside of the best practices as described in the **Guide to the Business Architecture Body of Knowledge**. Such models may or may not be useful and valuable to the business architect; practice and results will determine this validity and may result in additions to the **Guide to the Business Architecture Body of Knowledge**. The models that may be constructed from the meta-model should be guided by the practices recommended in the Guide.

However, the meta-model should prohibit, whenever possible, the construction of a model that specifically conflicts with the principles established in the **Guide to the Business Architecture Body of Knowledge**.

